EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 7

1. Let f be a probability function for a distribution. Suppose that $f(x) = 0$ for $x \notin [0, 1]$. Prove that the variance of this distribution is at most $1/4$. Next, prove that this bound is tight by providing a distribution which achieves the upper bound.

2. Let X_n be a random variable with the binomial distribution $\text{Bin}(n, p_n)$. Assume that $\lim_{n\to\infty} np_n = \lambda \in [0, \infty)$. Prove that the moment generating function (m.g.f.) of X_n converges pointwise (i.e. for every real t) to the m.g.f. of the Poisson distribution with mean λ.

3. Let $X_1, X_2, ...$ be i.i.d positive random variables with mean $a > 0$. Let $Y_1, Y_2, ...$ be i.i.d positive random variables with mean $b > 0$. Show that

$$\frac{X_1 + X_2 + ... + X_n}{Y_1 + Y_2 + ... + Y_n} \to c$$

where $c > 0$, with probability 1. What is the value of c? Does it matter whether the $X_i's$ are independent of the $Y_j's$?

4. Let $U_1, U_2, ..., U_{60}$ be i.i.d Unif(0, 1), and $X = U_1 + U_2 + ... + U_{60}$. Approximate the distribution of X with appropriate parameters, and give a simple but accurate approximation for $P(X > 17)$. Justify your answer.

5. (a) Consider i.i.d. Poisson(λ) r.v.s $X_1, X_2, \ldots$. The MGF of X_j is $M(t) = e^{\lambda(e^t-1)}$. Find the MGF $M_n(t)$ of the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j.$$

(b) Find the limit of $M_n(t)$ as $n \to \infty$.

6. (a) Suppose you toss a fair coin 100 times. Find the probability that you end up with at least 90 heads using Markov inequality.
(b) Can you come up with a non-negative random variable X with $E[X] = 50$ that achieves the upper bound above for $P(X > 90)$?
(c) For a general non negative random variable X that has a finite expectation, show that the equality case can always be achieved in the Markov Inequality. By doing so, you are showing that Markov Inequality cannot be improved in general.

7. Let $X_1, X_2, \ldots, X_n$ be a sequence of i.i.d. random variables with mean $\mu = 10$ and variance $\sigma^2 = 25$. Define $S_n = \sum_{i=1}^{n} X_i$ and $\overline{X}_n = \frac{S_n}{n}$.

1. According to the Central Limit Theorem, what is the distribution of $\overline{X}_n$ as $n \to \infty$?
2. For $n = 100$, calculate the approximate probability $P(9.5 \leq \overline{X}_n \leq 10.5)$.

8. The total number of students attending class on any given week is a random variable with mean 50. If the variance in a week's attendance is known to be 25, then the probability that the attendance in any week will be between 40 and 60 is at least _____ ?

9. A physicist makes 25 independent measurements of the specific gravity of a certain body. He knows that the limitations of his equipment are such that the standard deviation of each measurement is σ units.

(a) By using the Chebyshev inequality, find a lower bound for the probability that the average of his measurements will differ from the actual specific gravity of the body by less than $\sigma/4$ units.

(b) By using the central limit theorem, find an approximate value for the probability in part (a).

1. Let f be a probability function for a distribution. Suppose that $f(x) = 0$ for $x \notin [0, 1]$. Prove that the variance of this distribution is at most $1/4$. Next, prove that this bound is tight by providing a distribution which achieves the upper bound.

$x^2 \leq x$ for $x \in [0,1]$

$\Rightarrow E[x^2] \leq E[x]$

$E(x^2] - (E[x])^2 \leq E[x] - (E[x])^2 \leq 1/4$

$Var(x) = 1/4 \Rightarrow E[x^2] - (E[x])^2 = 1/4$

$\Rightarrow E[x^2] = E[x] = 0$ or 1

x is discrete 0 and 1 (Bernoulli)

$P(x=0) = p \quad P(x=1) = 1-p$

$E[x] = 1/2 \Rightarrow p = 1/2$ (coin flip)

8. The total number of students attending class on any given week is a random variable with mean 50. If the variance in a week's attendance is known to be 25, then the probability that the attendance in any week will be between 40 and 60 is at least _____ ?

$\mu = 50 \quad P(50-10 \leq x \leq 50+10) \quad \sigma = \sqrt{25} = 5$

$= P(|x-50| \leq 10)$

$P(|x-\mu| \leq k\sigma) > 1 - 1/k^2 \quad k=2 \Rightarrow P \geq \frac{3}{4}$

6. (a) Suppose you toss a fair coin 100 times. Find the probability that you end up with at least 90 heads using Markov inequality.
(b) Can you come up with a non-negative random variable X with $E[X] = 50$ that achieves the upper bound above for $P(X > 90)$?
(c) For a general non negative random variable X that has a finite expectation, show that the equality case can always be achieved in the Markov Inequality. By doing so, you are showing that Markov Inequality cannot be improved in general.

(a) $P(x \geq 90) \leq \frac{E[x]}{90} = \frac{100 \cdot 1/2}{90} = 5/9$

(b) $P(x=90) = 5/9 \quad P(x=0) = 4/9$

(c) $x = 0$ and a $\qquad P(x \geq a) \leq \frac{E[x]}{a}$

$P(x=a)$ is μ/a

4. Let $U_1, U_2, ..., U_{60}$ be i.i.d Unif(0, 1), and $X = U_1 + U_2 + ... + U_{60}$. Approximate the distribution of X with appropriate parameters, and give a simple but accurate approximation for $P(X > 17)$. Justify your answer.

→ according to CLT

$X_i \sim N(\mu_x, \sigma_x^2)$ $\mu = 60 \cdot 1/2 = 30$ $\sigma^2 = 60 \times \frac{1}{12} =$

$\mu_i = \frac{1}{2}$, $\sigma_i = \frac{1}{12}$

$X \sim N(30, 5)$

$P(X > 17) = P(Z > \frac{17-30}{\sqrt{5}})$

$P(Z < 5.81)$

$= \Phi(5.81)$

9. A physicist makes 25 independent measurements of the specific gravity of a certain body. He knows that the limitations of his equipment are such that the standard deviation of each measurement is σ units.

   (a) By using the Chebyshev inequality, find a lower bound for the probability that the average of his measurements will differ from the actual specific gravity of the body by less than $\sigma/4$ units.

   (b) By using the central limit theorem, find an approximate value for the probability in part (a).

(a) $E[X_i] = \mu$, $SD(X_i) = \sigma$

$\bar{X} = \frac{\sum X_i}{n} \Rightarrow E[\bar{X}] = \mu$; $Var[\bar{X}] = \frac{Var(X_i)}{n} = \frac{\sigma^2}{25}$

$\Rightarrow SD[\bar{X}] = \sigma/5$

$P[|\bar{X} - \mu| < \sigma/4) \geq 1 - 1/k^2$ $k = 5/4$

$\geq 9/25$

(b) $\bar{X} = N(\mu, \sigma^2/25)$

$P(|X - \mu| < \sigma/4)$

$= P(\frac{|X-\mu|}{\sigma/5} < \frac{\sigma/4}{\sigma/5}) = P(|Z| < 5/4)$

$= P(Z < 1.25) - P(Z < -1.25) = 2P(Z < 1.25) - 1$

$\because P(Z < -1.25) = 1 - P(Z < 1.25)$ $= 2\Phi(1.25) - 1$

7. Let $X_1, X_2, \ldots, X_n$ be a sequence of i.i.d. random variables with mean $\mu = 10$ and variance $\sigma^2 = 25$. Define $S_n = \sum_{i=1}^{n} X_i$ and $\overline{X}_n = \frac{S_n}{n}$.

   1. According to the Central Limit Theorem, what is the distribution of $\overline{X}_n$ as $n \to \infty$?
   2. For $n = 100$, calculate the approximate probability $P(9.5 \leq \overline{X}_n \leq 10.5)$.

1. $E[\bar{X}_n] = 10$, $\sigma[\bar{X}_n] = 5/\sqrt{n}$

$\bar{X}_n \approx N(10, 25/n)$

2. $\bar{X}_n \sim N(10, 0.25)$ $\quad \sigma = 0.5 \quad \mu = 10$

$P(9.5 \le \bar{X}_n \le 10.5) = P(|X - 10| \le 0.5)$

$= P(z \le 1) - P(z \le -1)$

$= \Phi(1) + \Phi(1) - 1 = 2\Phi(1) - 1$

2. Let X_n be a random variable with the binomial distribution $\mathrm{Bin}(n, p_n)$. Assume that $\lim_{n\to\infty} np_n = \lambda \in [0, \infty)$. Prove that the moment generating function (m.g.f.) of X_n converges pointwise (i.e. for every real t) to the m.g.f. of the Poisson distribution with mean λ.

$M_{X_n}(t) = E[e^{tX_n}] = \{e^0(1-p_n) + e^t[p_n]\}^n$

$= \{1 + p_n(e^t - 1)\}^n$

$M_Y(t) = E[e^{tY}] = \sum_{k=0}^{\infty} e^{tk} \cdot \frac{e^{-\lambda}\lambda^k}{k!} = e^{\lambda(e^t - 1)}$

$(1+x)^n \approx e^{nx}$ for large n

$L = \lim_{n\to\infty} (1 + x/n)^n \to \ln(L) = n \ln\left(1 + \frac{x}{n}\right)$

$= \frac{\ln(1 + x/n)}{(1/n)}$

$\ln L \doteq \lim_{t\to 0} \frac{\ln(1+tx)}{t} = x \quad \Rightarrow L = e^x$

3. Let $X_1, X_2, \ldots$ be i.i.d positive random variables with mean $a > 0$. Let $Y_1, Y_2, \ldots$ be i.i.d positive random variables with mean $b > 0$. Show that

$$\frac{X_1 + X_2 + \ldots + X_n}{Y_1 + Y_2 + \ldots + Y_n} \to c$$

where $c > 0$, with probability 1. What is the value of c? Does it matter whether the $X_i's$ are independent of the $Y_j's$?

$\lim_{n\to\infty} \bar{X_n} = a$ $\lim_{n\to\infty} \bar{Y_n} = b \Rightarrow c = a/b$, Independence not required

5. (a) Consider i.i.d. Poisson(λ) r.v.s $X_1, X_2, \ldots$. The MGF of X_j is $M(t) = e^{\lambda(e^t-1)}$. Find the MGF $M_n(t)$ of the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j.$$

(b) Find the limit of $M_n(t)$ as $n \to \infty$.

$E[e^{t/n X_1}] \cdot E[e^{t/n X_2}] \ldots = M(t/n) \ldots M(t/n)$

$= e^{\lambda n(e^{t/n}-1)}$

$\lim_{n\to\infty} \lambda n(e^{t/n}-1)$ $1/n = u \Rightarrow \lim_{u\to 0} \frac{\lambda(e^{tu}-1)}{u} = \lambda t$

$\Rightarrow \lim_{n\to\infty} M_n(t) = e^{\lambda t}$

11. Let $X_1, \ldots, X_n$ be independent random variables with $X_i \sim \text{Uniform}(0,1)$. Define

$$\widehat{I}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i), \qquad g(x) = e^x.$$

Let $I = \mathbb{E}[g(X)]$. Find a sample size n such that

$$\mathbb{P}\big(|\widehat{I}_n - I| < 0.01\big) \geq 0.95.$$

$\text{Var}(g(x)) = E[e^{2x}] - (E[e^x])^2$

$\int_0^1 e^{2x}\cdot dx = \frac{e^2-1}{2}$ $\int_0^1 e^x \cdot 1 \cdot dx = (e-1)$

$g(x_i) = e^{x_i}$ $E[\widehat{I}_n] = E[e^{x_i}] = I$

$\sigma = \text{Var}(y_i) = \frac{\text{Var}(e^{x_i})}{n} \Rightarrow \widehat{I}_n - I \sim N(0, \sigma)$

$2\sigma = 0.01 \Rightarrow 2\cdot\frac{\sigma_y}{\sqrt{n}} = 0.01$

$4\cdot\sigma_y^2 = 10^{-4}\cdot n \Rightarrow n = 4\times 10^4 \sigma_y^2 = 9681.2$

$\sigma_y^2 = \frac{e^2-1}{2} - (e-1)^2 = \frac{e^2-1}{2} - e^2 + 2e - 1 = -\frac{e^2}{2} + 2e - \frac{3}{2} = 0.085$

$n \sim 9682$

10. A political candidate is concerned about voter support in a large district. Her campaign manager claims that she has 52% support ($p = 0.52$). The candidate will only proceed with a new, expensive advertising campaign if she is convinced her support is less than 50%. They decide to conduct a poll of n voters. They will launch the new campaign if the sample proportion of supporters, $\hat{p}$, is less than 50%. Assuming the campaign manager is correct (that the true support is $p = 0.52$), what is the **minimum number of voters** n they must poll to ensure that the probability of *incorrectly* launching the new campaign (i.e., finding $\hat{p} < 0.50$) is **no more than** 1%?

    Use the CLT to approximate probabilities as necessary. The critical value Z corresponding to a 1% tail probability, $P(Z \leq z) = 0.01$, is $\mathbf{z_{crit}} \approx \mathbf{-2.326}$.

11. Let $X_1, \ldots, X_n$ be independent random variables with $X_i \sim \text{Uniform}(0, 1)$. Define

$$\widehat{I}_n = \frac{1}{n} \sum_{i=1}^{n} g(X_i), \qquad g(x) = e^x.$$

    Let $I = \mathbb{E}[g(X)]$. Find a sample size n such that

$$\mathbb{P}\big(|\widehat{I}_n - I| < 0.01\big) \geq 0.95.$$

    (Hint: For a normal variable $Z \sim \mathcal{N}(0, \sigma^2)$, $\mathbb{P}(|Z| < 2\sigma) = 0.95$)

12. Suppose $X_1, \ldots, X_n \sim$ iid (continuous) Uniform$[0, 1]$. Let

$$Y_n = e^{X_1} + \cdots + e^{X_n}.$$

    1. What is the range of values taken by Y_n?
    2. Find $E[Y_n]$ and $\text{Var}(Y_n)$.
    3. Approximate $P(|Y_n - E[Y_n]| > \dfrac{n}{2})$ using Chebyshev's inequality.
    4. Approximate $P(Y_n > 2n)$ using the CLT.

12. Suppose $X_1, \ldots, X_n \sim$ iid (continuous) Uniform$[0, 1]$. Let

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4. Approximate $P(Y_n > 2n)$ using the CLT.

1. $e^{X_i} \in [1, e] \Rightarrow Y_n \in [n, ne]$

2. $E[Y_n] = E[e^{X_1} + e^{X_2} + \ldots] = n \cdot E[e^{X_i}] = n \int_0^1 e^x \, dx$

$= n(e-1)$

$E[Y_n^2] = n\left(\frac{1}{2}(e^2-1)\right)$

$\text{Var}(Y_n) = n\left[\frac{e^2}{2} - \frac{1}{2} - e^2 + 2e - 1\right]$

$= n\left[-\frac{e^2}{2} + 2e - \frac{3}{2}\right]$

3. $P(|Y_n - E[Y_n]| > k) \leq \frac{n\left[-\frac{e^2}{2} + 2e - \frac{3}{2}\right]}{n^2} \leq \frac{8e - 2e^2 - 6}{n}$

4. $Y_n \sim N(E[Y_n], \text{Var}(Y_n))$

$P(Y_n > 2n) = P\left(\frac{Y_n - n(e-1)}{\sqrt{(8e-2e^2-6)n}} > \frac{2n - n(e-1)}{\sqrt{(8e-2e^2-6)n}}\right)$

$= P\left(Z > \frac{\sqrt{n}(3-e)}{\sqrt{2e - e^2/2 - 3/2}}\right)$

$n \to \infty \Rightarrow 0$

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Use the CLT to approximate probabilities as necessary. The critical value Z corresponding to a 1% tail probability, $P(Z \leq z) = 0.01$, is $\mathbf{z_{crit}} \approx \mathbf{-2.326}$.

$\mu = 0.52 \quad \sigma_{\hat{p}} = \sqrt{\frac{0.52(0.48)}{n}} = \sqrt{\frac{0.2496}{n}}$

$P(\hat{p} < 0.5) \leq 0.01 \Rightarrow P\left(Z \leq \frac{-0.02}{\sqrt{\frac{2496}{10^4 n}}}\right) \leq 0.01$

$$\Rightarrow \frac{-0.02}{\sqrt{\frac{2496}{104n}}} \leq -2.326 \Rightarrow n \geq 3375.99 \Rightarrow n = 337$$